

Data Visualization Project: Florida Airports Case study

Orlando International Airport (MCO) vs Miami International Airport (MIA)

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Outline

- 1 Introduction - EDA
- 2 Data Analysis - A Deep Dive in Delay Times
- 3 Geography, Weather, and Market Power
- 4 EDA & Performance Tool
- 5 The Bigger Picture - Conclusions

- **Objective:** Compare operational patterns of Orlando (MCO) and Miami (MIA) airports
- **Dataset:** U.S. Department of Transportation flight data (1.59M flights)
- **Time Period:** 2004-2008 (2008 partial: Jan-Apr only)
- **Key Questions:**
 - How do traffic volumes compare between MCO and MIA?
 - What are the key operational differences?
 - How do temporal patterns differ between the airports?

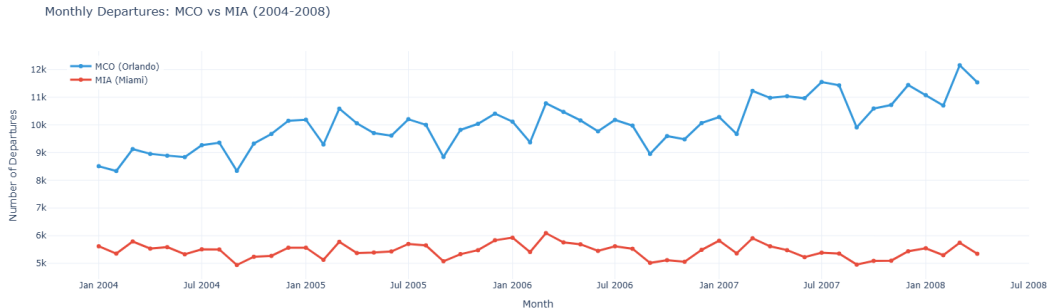
Data Processing:

- Filtered 1.59M flights for MCO/MIA
- Created derived features (seasonality, time-of-day)
- Handled missing values and cancellations

Visualization Tools:

- Python + Plotly (interactive charts)
- Jupyter Notebooks
- Time-series, heatmaps, comparisons

Monthly Departures: MCO vs MIA



Key Observations:

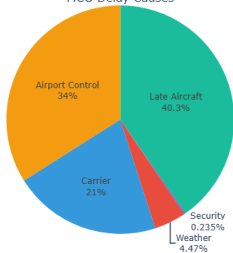
- MCO handles 2x more departures — 10,000 monthly vs 5,000
- Summer peaks at both airports — June-August highest traffic

Causes of Delay

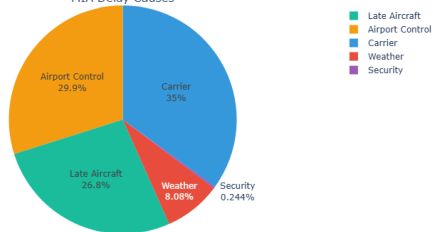
Primary Causes of Delays (2004-2008)

NAS = National Aviation System (Air Traffic Control, Airport Operations)

MCO Delay Causes



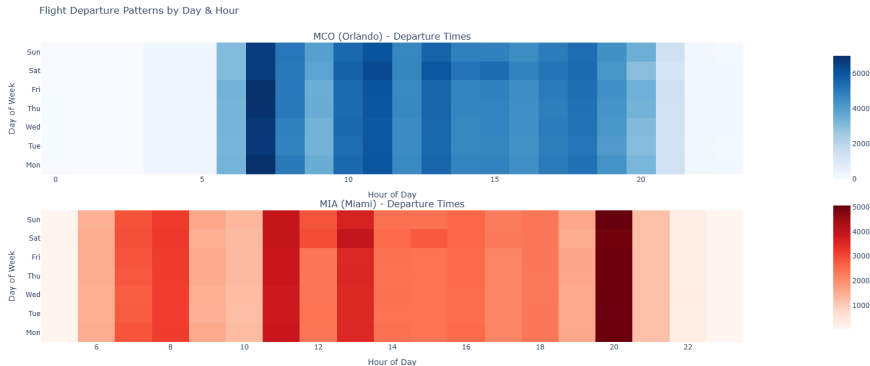
MIA Delay Causes



Key Observations:

- Late Aircraft is 1 cause — 40% at both airports (cascading delays)
- NAS (Air Traffic Control) is 2 — 34-35% system capacity issues

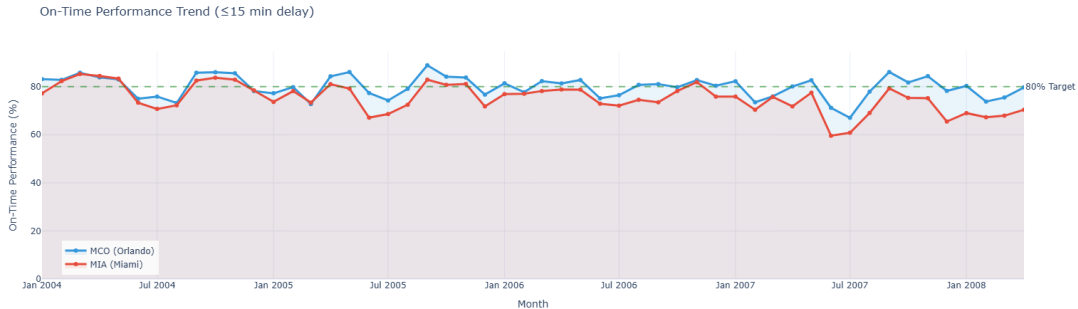
Departure Time Patterns (Day & Hour)



Key Observations:

- MCO peaks at 7 AM then evenly distributed—Orlando shows consistent traffic throughout the day
- MIA has three distinct peaks — 8 AM, 11 AM, and 8 PM show concentrated activity
- Both airports show less variation on Sat-Sun

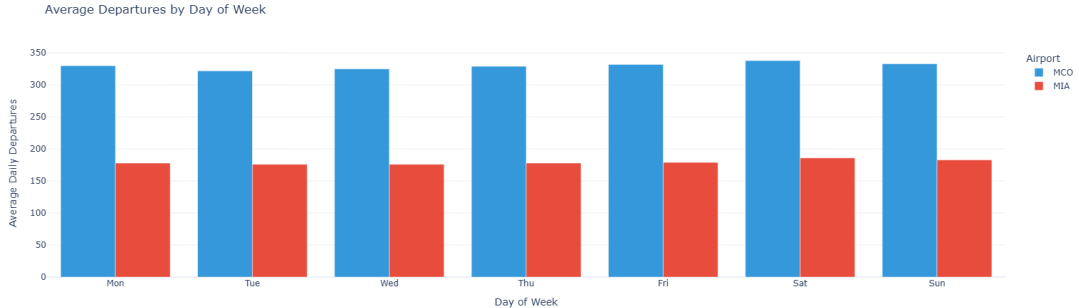
On-Time Performance Trend



Key Observations:

- MCO outperforms MIA — 79% OTP vs 74%
- Both decline 2004-2007 — System-wide congestion visible

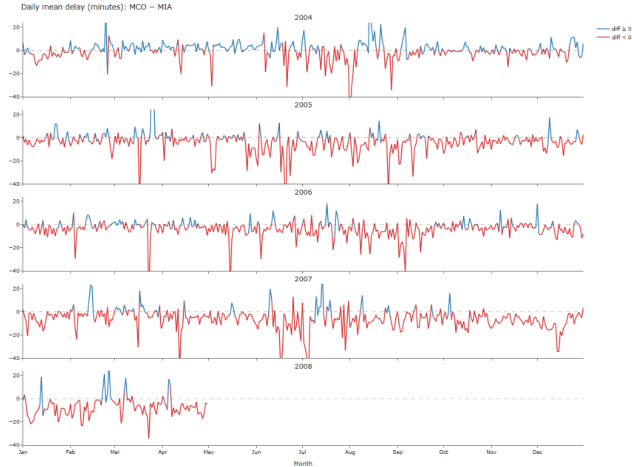
Weekly Traffic Patterns



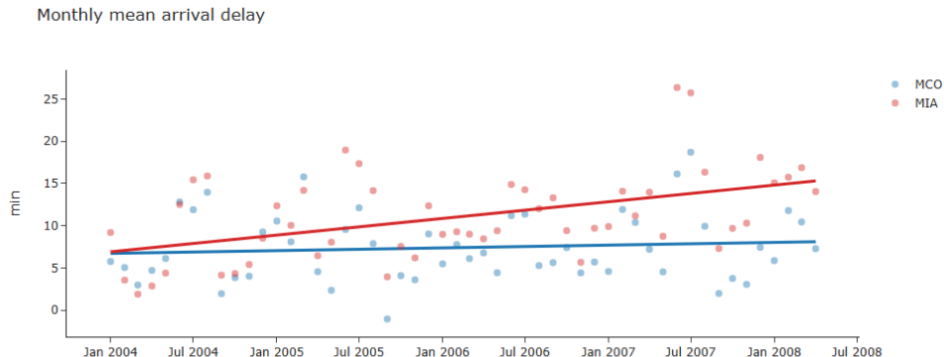
Key Observations:

- Weekend is actually the busiest — Saturday highest, followed by Sunday and Friday
- Weekdays show consistent baseline — Mon-Thu maintain similar lower traffic levels

Deep Dive in Delays: Daily Mean Gap (MCO – MIA), 2004–2008



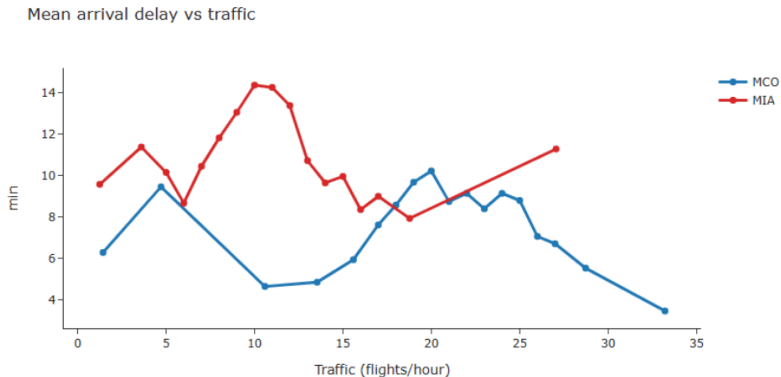
Long-Term Trend Divergence: Monthly Mean Arrival Delay



Key Observations:

- **MCO:** Flat trend ($\approx +0.03$ min/month, not significant) — stable, mean-reverting performance.
- **MIA:** Upward trend ($\approx +0.16$ min/month, $p < 0.001$) — persistent, compounding delay growth.

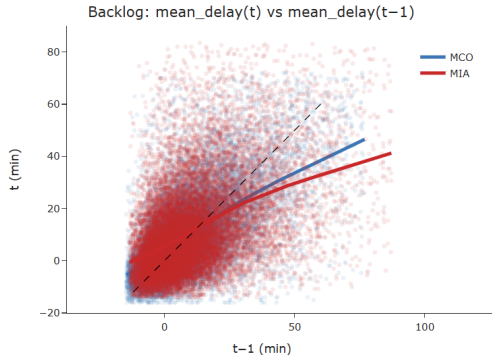
Traffic Sensitivity: Where Delays Break (MCO vs MIA)



Key Observations:

- **MCO:** Near-linear, predictable scaling of delay with traffic.
- **MIA:** Nonlinear "danger zone" at ~ 10 – 15 flights/hour; small traffic increases $\rightarrow$ large delay jumps.

Delay Backlog: Current vs Previous Hour

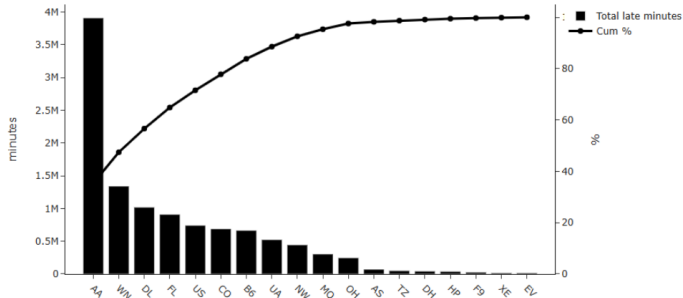


Key Observations:

- Both airports lie below the $y = x$ line: delays partially carry over hour-to-hour.
- MCO shows stronger persistence (steeper LOWESS); MIA resets more frequently.

Pareto of System Pain: Who Contributes to Delay?

Pareto: late-arrival minutes by airline



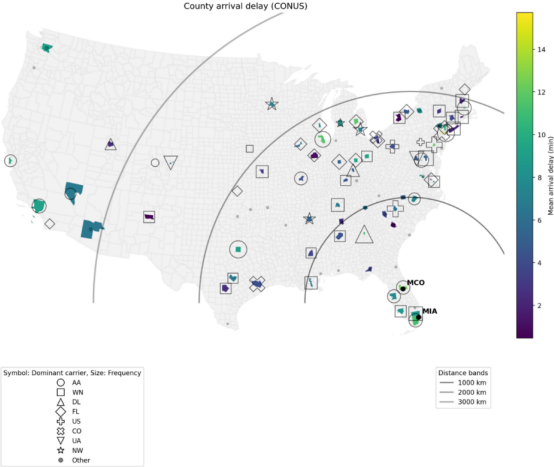
Key Insight:

- Delay minutes are **highly concentrated** among a small number of carriers.
- **American Airlines alone accounts for over one-third** of total delay minutes.
- Given AA's dominance at MIA, this creates a **carrier-airport coupling** that amplifies congestion.

Implication:

- This is not just an airline issue, but a **hub governance** problem.

Spatial Distribution of Arrival Delays (CONUS)



- **Performance-Adjusted Landing Fees (PALF)**

- 15–20% surcharge for carriers in the bottom decile of hub-specific on-time performance.
- Internalizes congestion externalities; targets chronic contributors.

- **Slot Recapture and Competitive Pressure**

- Trigger divestiture if a carrier controls $>50\%$ of slots and contributes $>30\%$ of delay minutes.
- Reallocating as little as 5% of peak slots can materially reduce systemic risk.

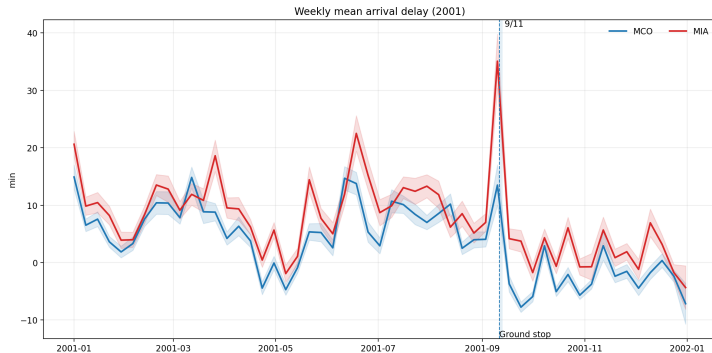
- **Time-of-Use Congestion Pricing**

- July evenings (18:00–21:00) identified as failure zones.
- 3× landing fees flatten demand, shift schedules, and reduce peak backlog formation.

- **Summer Resilience Buffers**

- Mandate 20% longer turnarounds for long-haul arrivals in summer.
- Invest in high-speed taxi exits and A-CDM systems.

Weekly Mean Arrival Delay (2001): Pre/Post 9/11



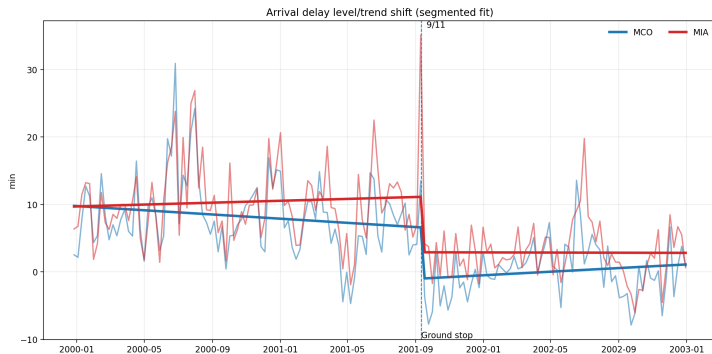
Key Observations:

- Sharp break at 9/11, followed by lower weekly mean delays.
- Reduced volatility in the post period.

Interpretation:

- Delays improve for **completed flights**.
- Delay KPIs alone do not capture overall disruption.

Interrupted Time Series: Level Shift at 9/11



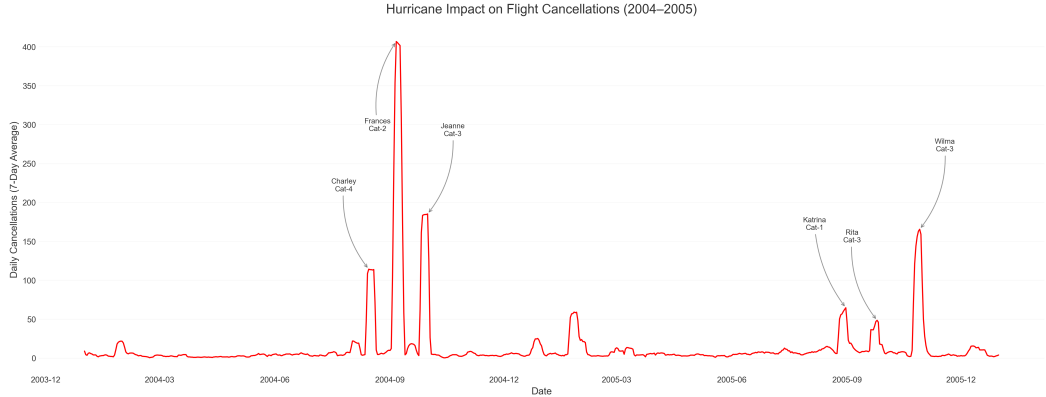
Key Observations:

- Clear step-down in delays post-9/11.
- Post-event trends are flat relative to the discontinuity.

Interpretation:

- Dominant effect is an **immediate level shift**.
- No evidence of a distinct MIA/MCO-specific response.

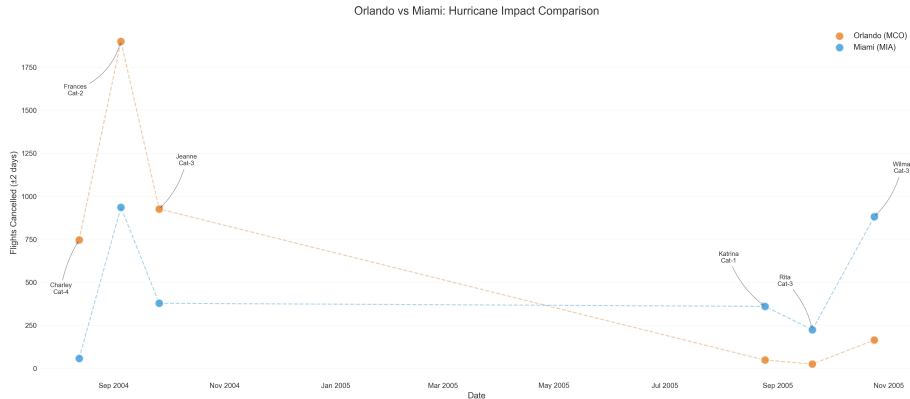
Hurricane Impact: Timeline of Disruption (2004–2005)



Key Observations:

- Hurricane Frances caused worst single-day impact — 400+ daily cancellations
- 2004-2005 saw 7 major named storms affecting Florida airports
- Storm trajectory and timing matter more than intensity alone

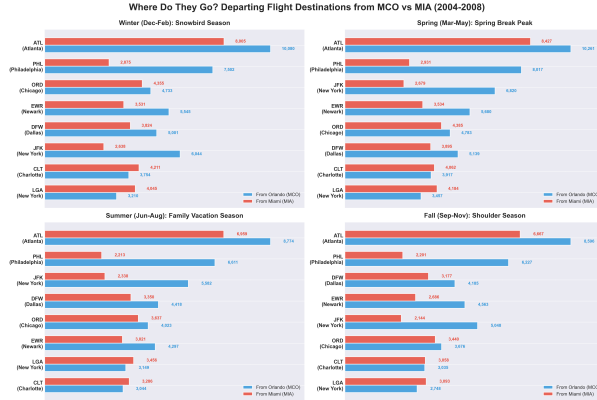
Geographic Vulnerability: MCO vs MIA Hurricane Response



Key Observations:

- MCO devastated in 2004 (Frances: 1,900 cancellations) — Central Florida direct hits
- MIA harder hit in 2005 (Katrina, Rita, Wilma) — Coastal exposure different risk
- Geographic positioning creates differential vulnerability to hurricane trajectories

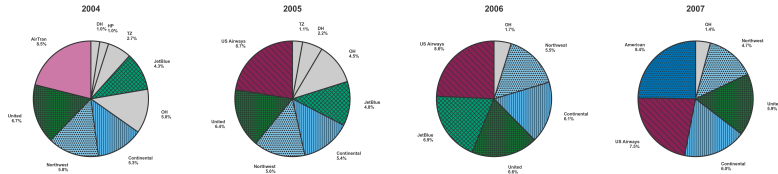
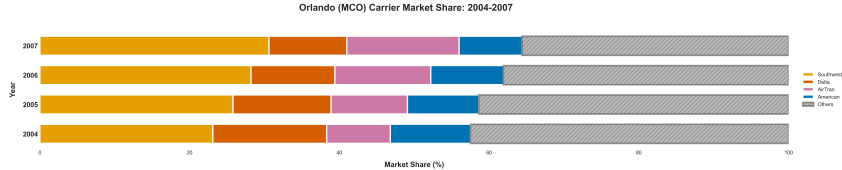
Route Profiles: Geography Shapes Destination Mix



Key Observations:

- MCO serves domestic hubs — ATL, ORD, DFW dominate (theme park tourism)
- MIA focuses on Northeast corridor — JFK, LGA, EWR (gateway to Latin America)
- Geographic positioning determines route structure and seasonal patterns

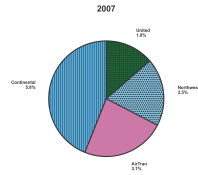
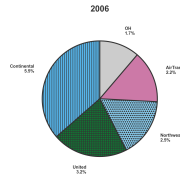
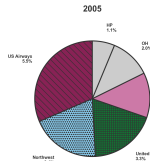
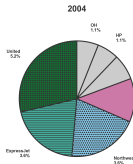
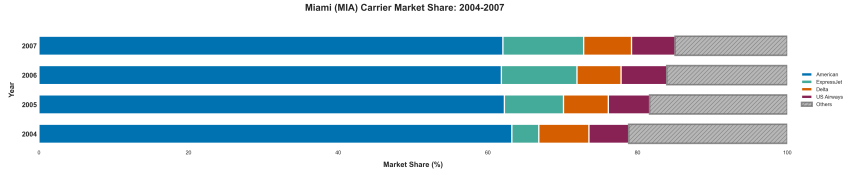
MCO Market Structure: Diversified Competition



Key Observations:

- No single carrier dominates MCO — Balanced competition across multiple airlines
- Domestic tourism market attracts diverse carriers (Southwest, Delta, others)
- Market share evolves year-to-year — Competitive environment prevents monopolization

MIA Market Structure: American Airlines Dominance



Key Observations:

- American Airlines controls 60%+ of MIA market — Stable dominance 2004-2007
- MIA serves as American's Latin America hub — Hub-and-spoke creates concentration
- International gateway routes favor incumbent dominance over competition

Summary: Geography Shapes Everything

Orlando (MCO):

- **Weather Risk:** Episodic shocks
- **Role:** Final destination
- **Market:** Competitive, multi-carrier

Result: Stable but weather-exposed

Miami (MIA):

- **Weather Risk:** Chronic pressure
- **Role:** International connector
- **Market:** Concentrated (AA 60%)

Result: Delay-amplifying structure

Geographic positioning determines risk profile, operational role, and market structure.

Traffic Volume: MCO vs MIA

Similar Punctuality, Divergent Risk: A Comparative Operational Analysis of MCO and MIA (01/01/2004 - 30/04/2008)

MCO handles twice the traffic	MCO exhibits lower cancellation rate despite traffic	MIA shows greater risk with more delayed flights	MIA outperforms MCO regarding On Time precision.	Different Risks are caused by different reasons	Distributional Statistics show greater risk for MIA.	MCO handles better busiest paths with lower delay.	Both Airports are affected by similar seasonal stress	Early departure hours show highest delay for both airports.
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For the same period of analysis (01/01/2004 - 30/04/2008), MCO handled nearly twice the total number of flights.



MIA Summary:
Airport type Public
Owner/Operator Miami-Dade Aviation Department
Serves Miami metropolitan area
Location Miami-Dade County, Florida, U.S.
Opened 1928; 98 years ago
Hub for ABX Air
Focus city for Avianca
Operating base for Frontier Airlines
Time zone EST (UTC-05:00)
- Summer (DST) EDT (UTC-04:00)
Elevation AMSL 3 m / 9 ft
Coordinates 25°47'36"N 080°17'26"W
Website miami-airport.com

MCO Summary:
Airport type Public
Owner/Operator Greater Orlando Aviation Authority
Serves Greater Orlando
Location Orlando, Florida, U.S.
Opened 1961; 65 years ago
Focus city for JetBlue
Operating base for Breeze Airways[1]
Built 1940; 86 years ago
Elevation AMSL 96 ft / 29 m
Coordinates 28°25'46"N 81°18'32"W
Website flymco.com

Cancellation Rates by Cause

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MCO handles twice the traffic

MCO exhibits lower cancellation rate despite traffic

MIA shows greater risk with more delayed flights

MIA outperforms MCO regarding On Time precision.

Different Risks are caused by different reasons

Distributional Statistics show greater risk for MIA.

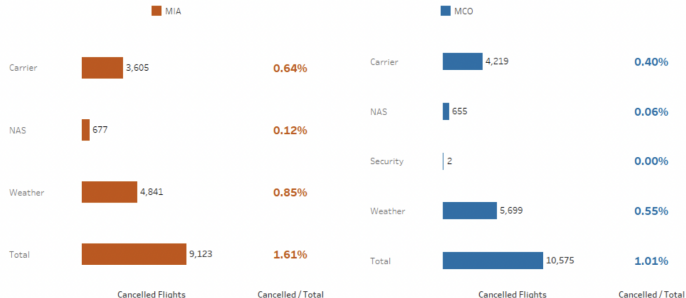
MCO handles better busiest paths with lower delay.

Both Airports are affected by similar seasonal stress

Early departure hours show highest delay for both airports.

Higher throughput does not inherently imply worse outcomes; MCO shows better handling regarding cancellation rates.

Despite handling significantly more flights overall, MCO exhibits a lower cancellation rate (1.01%) than MIA (1.61%), indicating lower relative operational reliability. Primary reason of cancellation for both airports is Weather.



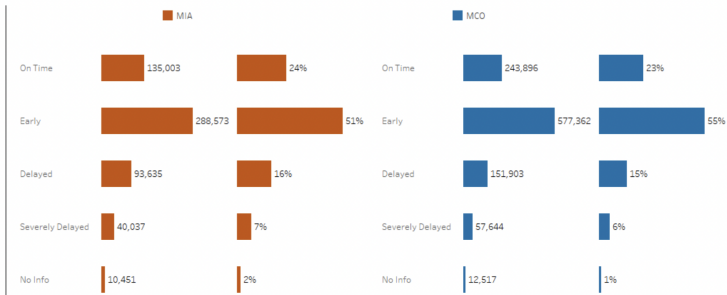
Flight Delay Categories

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Despite differences in traffic, punctuality is lacking for both airports with MIA showing higher risk.

Early flights are dominant for both airports showing lack of punctuality. On Time precision is nearly identical. Regarding delays, MIA shows 1p.p. difference both in Delayed and Severely Delayed flights indicating higher risk.

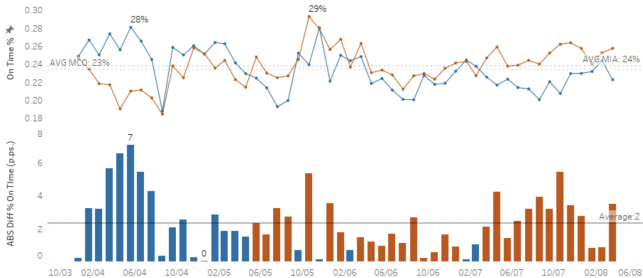


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MIA demonstrates higher reliability dominating in monthly On Time precision despite MCO better results in early period.

MIA outperforms MCO regarding On Time flights as indicated both by the higher average as well as the number of months with superior results. **Common external factors** affect the two airports as volatility shows similar fluctuations for both airports. **Largest performance gaps are present in the early stages** with later periods showing more stable differences and an **average gap of approximately 2 percentage points.** (Excludes flights regarding MCO-MIA or MIA-MCO)



Primary Delay Causes

Similar Punctuality, Divergent Risk: A Comparative Operational Analysis of MCO and MIA (01/01/2004 - 30/04/2008)

MCO handles twice the traffic

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Different Risks are caused by different reasons

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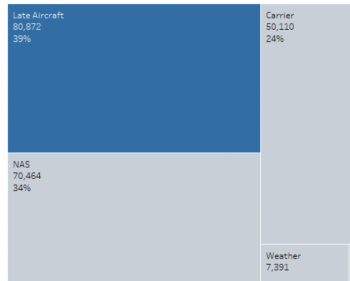
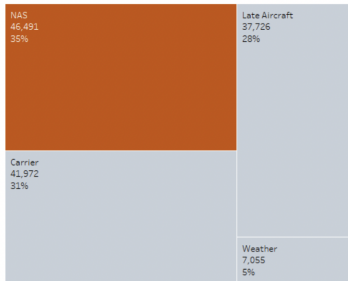
MCO handles better busiest paths with lower delay.

Both Airports are affected by similar seasonal stress

Early departure hours show highest delay for both airports.

The two airports have different reasons for failure; MCO snowballs while MIA is prone to external dependencies.

Primary reason for delay (delayed & severely delayed) differs with NAS being the most significant reason for MIA while Late Aircraft is the most significant for MCO. Delays at MCO are more likely to snowball while MIA is more prone to external dependencies.



Similar Punctuality, Divergent Risk: A Comparative Operational Analysis of MCO and MIA (01/01/2004 - 30/04/2008)

MCO exhibits lower cancellation rate despite traffic

MIA shows greater risk with more delayed flights

MIA outperforms MCO regarding On Time precision.

Different Risks are caused by different reasons

Distributional Statistics show greater risk for MIA.

MCO handles better busiest paths with lower delay.

Both Airports are affected by similar seasonal stress

Early departure hours show highest delay for both airports.

MIA schedule reliable & risk prone; MCO less delay and precision.

MIA exposes passengers to higher risks as is indicated by comparing the distributional statistics.

MIA shows greater risk regarding delays with higher total average delay. Higher total average is accompanied by greater variability with IQR being more than 10 minutes greater. Both medians are around zero which shows that more extreme values are present in MIA.

Avg. Total Delay	Percentile (25) of Total Delay	Median Total Delay	Percentile (75) of Total Delay	IQR Total Delay	Variance of Total Delay	Std. dev. of Total Delay
38	-11	0	37	48	13K	115
30	-12	-1	25	37	10K	100

Total delay is giving by summing the minutes of delay across participating categories.

Top 5 Busiest Routes Analysis

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Top 5 Busiest paths analysis backs MIA higher risk; MCO shows lower delays despite higher traffic.

MIA's highest-traffic routes are more delay-intensive on a per-flight basis than MCO's, while MCO's busiest paths carry higher volumes but relatively lower delay per flight, indicating stronger congestion and complexity effects at MIA's core routes.



Path	Direction of Path	Flights	✈	Total Delay	Total Delay / Flights
ATL-MIA	Incoming	30,063		988,063	33
MIA-ATL	Outgoing	30,058		1,092,426	36
ORD-MIA	Incoming	15,819		835,324	53
MIA-ORD	Outgoing	15,817		877,852	56
MIA-LGA	Outgoing	14,778		798,352	54

Path	Direction of Path	Flights	✈	Total Delay	Total Delay / Flights
MCO-ATL	Outgoing	37,711		1,223,217	32
ATL-MCO	Incoming	37,609		1,356,744	36
MCO-PHL	Outgoing	28,357		1,176,659	41
PHL-MCO	Incoming	28,242		1,288,638	46
JFK-MCO	Incoming	23,499		897,450	38

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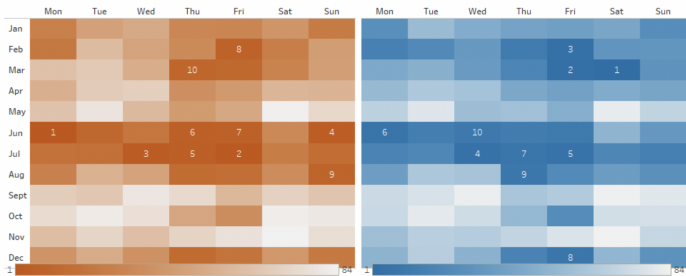
Both Airports are affected by similar seasonal stress

Early departure hours show highest delay for both airports.

MIA schedule reliable & risk prone; MCO less delay and precision.

Seasonality delay analysis shows that both airports are affected by seasonal stress with MCO also having extension to winter and spring.

Both airports share common delay pattern regarding the summer months which indicates shared seasonal stress period. Although MIA shows most high ranked delays in June and July, MCO shows extension to winter and spring, indicating a buildup of congestion rather than a sharper peak at summer.



Hourly Delay Analysis

Similar Punctuality, Divergent Risk: A Comparative Operational Analysis of MCO and MIA (01/01/2004 - 30/04/2008)

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Distributional Statistics show greater risk for MIA.

MCO handles better busiest paths with lower delay.

Both Airports are affected by similar seasonal stress

Early departure hours show highest delay for both airports.

MIA schedule reliable & risk prone; MCO less delay and precision.

Early departure hours show biggest delays while later in the day delays are generally evenly distributed with minor buildups.

MIA and MCO both show pronounced total-delay spikes in the 0–2 departure-hour window; beyond this period, MIA's delays are relatively evenly distributed, while MCO exhibits a modest buildup in delays leading into the early-hour peak.

Departure time radial plots indicating avg total delay using bar height.



Similar Punctuality, Divergent Risk: A Comparative Operational Analysis of MCO and MIA (01/01/2004 - 30/04/2008)

MCO exhibits lower cancellation rate despite traffic	MIA shows greater risk with more delayed flights	MIA outperforms MCO regarding On Time precision.	Different Risks are caused by different reasons	Distributional Statistics show greater risk for MIA.	MCO handles better busiest paths with lower delay.	Both Airports are affected by similar seasonal stress	Early departure hours show highest delay for both airports.	MIA schedule reliable & risk prone; MCO less delay and precision.
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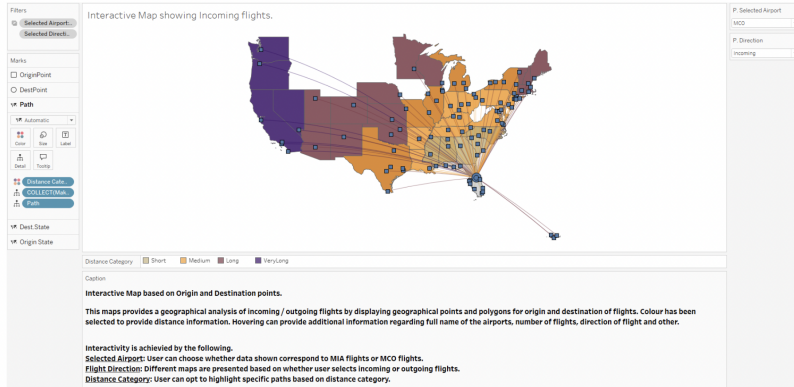
Overall Outcome

Miami International Airport (MIA) demonstrates higher on-time consistency over time but exhibits greater delay variability, higher tail risk, and stronger sensitivity to network and external constraints.

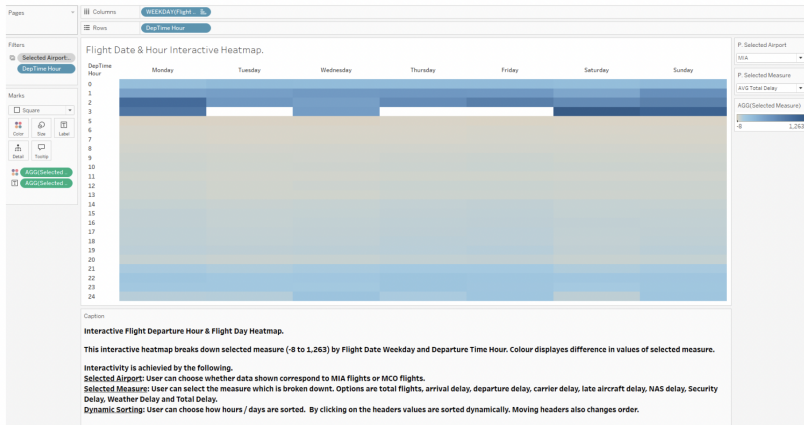
Orlando International Airport (MCO), despite operating at nearly double the traffic volume, shows more predictable delay behavior, lower severity risk, and more stable performance across routes and hours of the day.

As a result, MIA can be characterized as schedule-reliable but risk-prone, while MCO is less dominant in on-time leadership but more operationally stable and resilient.

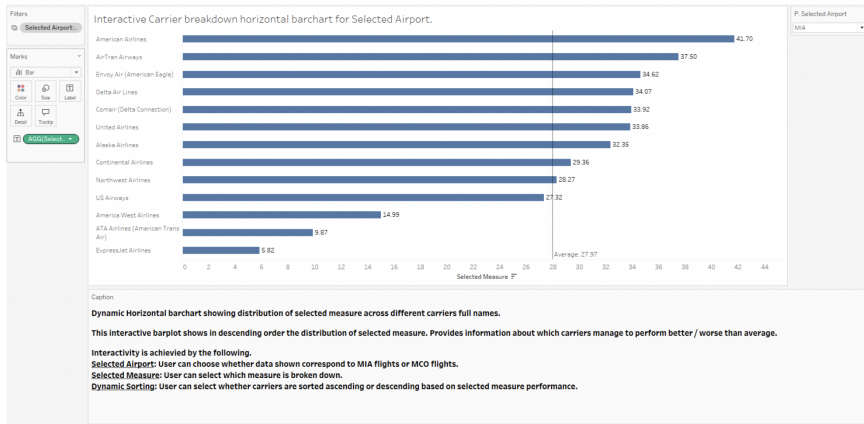
Interactive Plots: Dynamic Map showing flights Geography



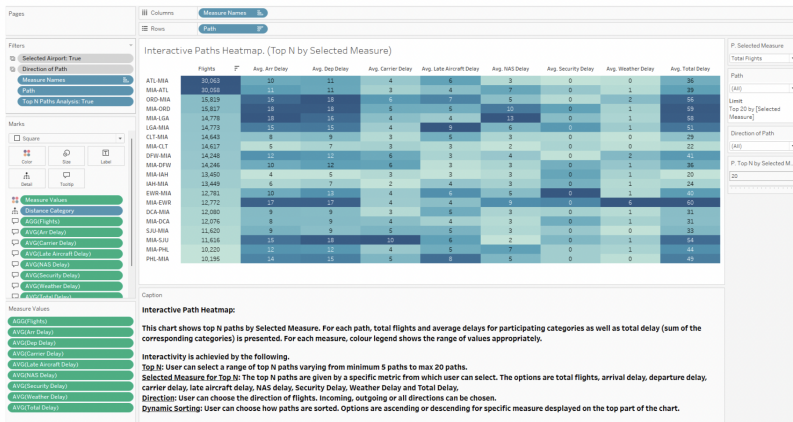
Interactive Plots: Dynamic Day & Hour Heatmap.



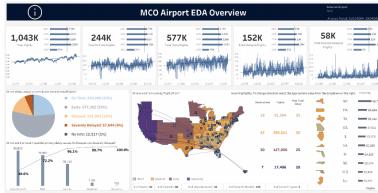
Interactive Plots: Dynamic Carrier Horizontal Barplot



Interactive Plots: Dynamic Paths Heatmap.



Interactive Plots: Dynamic Map showing flights Geography



Key Findings Summary

- ➊ **Finding 1:** MCO demonstrates stable, predictable operations while MIA shows fragile, congestion-prone behavior
- ➋ **Finding 2:** Hub-and-spoke model at MIA creates carrier concentration and delay amplification
- ➌ **Finding 3:** Geographic positioning drives differential hurricane vulnerability and route structures
- ➍ **Finding 4:** Weather acts as trigger, but recovery capacity determines operational outcomes

Implications:

- Airport function (leisure vs hub) fundamentally shapes operational resilience
- Policy interventions must target structural causes, not symptoms

Limitations:

- 2008 data incomplete (Jan-Apr only)
- Focus on delays over cancellations
- Historical data (2004-2008 period)

Future Directions:

- Extended temporal analysis post-2008
- Integrated delay-cancellation metrics
- Real-time predictive modeling

Thank You

Questions?

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